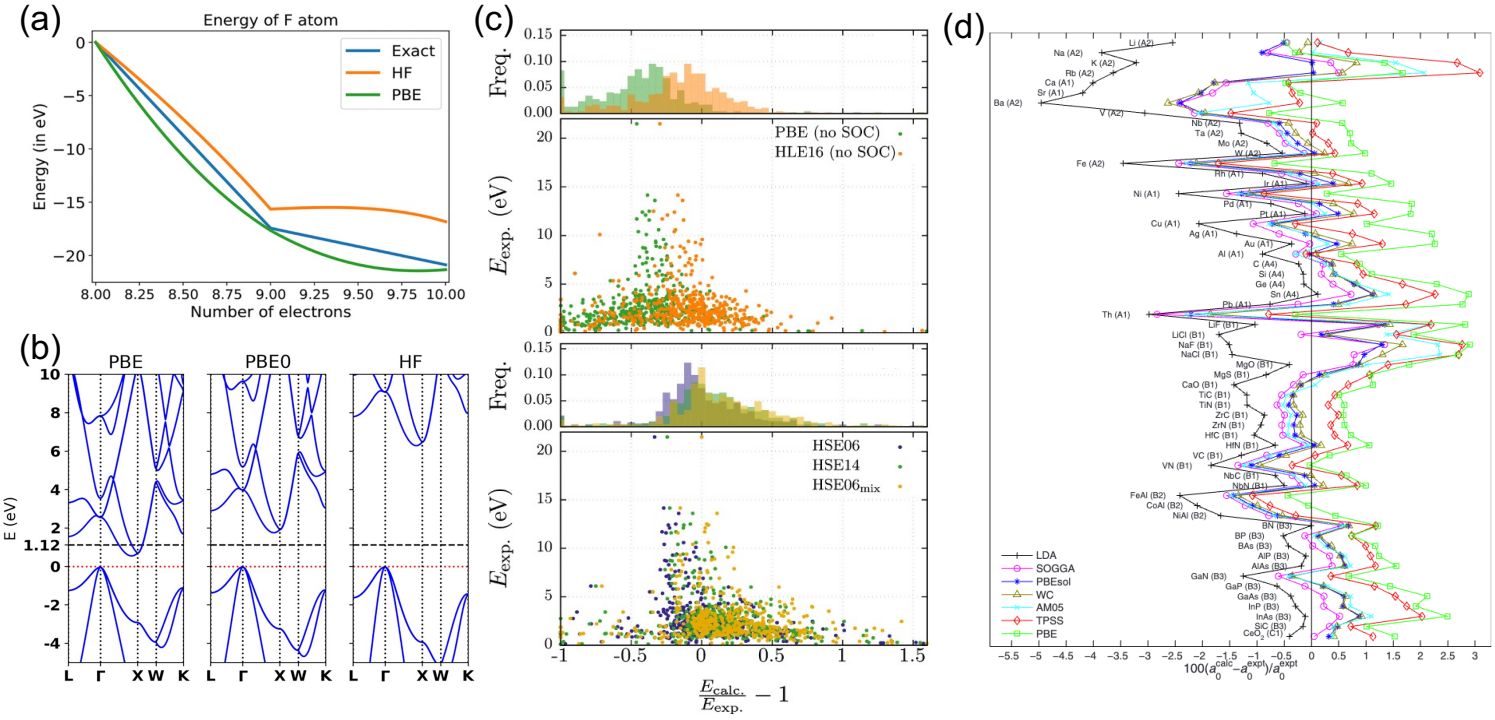

(a)
Energy of F atom
Exact
HF
PBE
Energy (in eV)
Number of electrons
(b)
PBE
PBE0
HF
E (eV)
1.12
L Γ X W K
(c)
Freq.
PBE (no SOC)
HLE16 (no SOC)
HSE06
HSE14
HSE06mix
Eexp. (eV)
Ecalc./Eexp. − 1
(d)
LDA
SOGGA
PBEsol
WC
AM05
TPSS
PBE
100(a0calc − a0expt)/a0expt